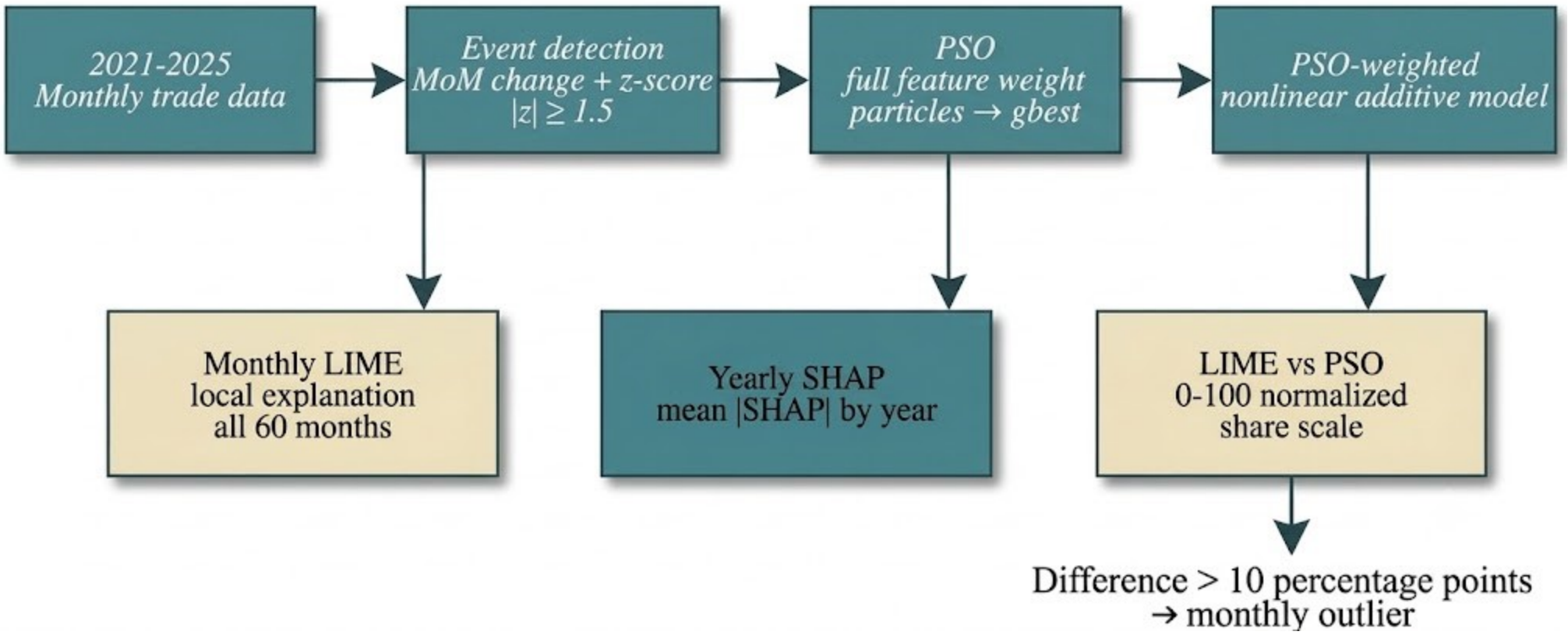

2021-2025
Monthly trade data
Event detection
MoM change + z-score
|z| ≥ 1.5
PSO
full feature weight
particles → gbest
PSO-weighted
nonlinear additive model
Monthly LIME
local explanation
all 60 months
Yearly SHAP
mean |SHAP| by year
LIME vs PSO
0-100 normalized
share scale
Difference > 10 percentage points
→ monthly outlier